Advanced ML Scenario Based Set - 1

1. **A healthcare startup wants to build a model to predict the onset of diabetes. The dataset has 50 features, but the team wants to use only the most relevant ones. How can they apply feature selection effectively?**

Answer:

Machine learning concept – Feature Selection

Steps:

- Collect data from patient health records (e.g., BMI, glucose, age, family history, etc.).
- Preprocess the data: handle missing values and scale features.
- Use correlation heatmaps to eliminate highly correlated features.
- Apply SelectKBest with mutual information to identify the top features.
- Optionally, apply Recursive Feature Elimination (RFE) with Logistic Regression.
- Train the model using selected feature
- Evaluate using Accuracy, ROC-AUC, and F1-Score

2. **You're creating a new project for managing student data. The project name is student_portal. What is the step-by-step command to start it? write logic for this django project**

Answer:
Steps:

- Open your terminal or command prompt.
- Navigate to the directory where you want your project.
- Run the command:
  django-admin startproject student_portal
- This creates a folder named student_portal with default Django files.

Advanced ML Scenario Based Set - 1

3. **A company is analyzing customer purchase patterns with 200+ behavioral features. How can they reduce dimensionality without losing predictive power?**

Answer:

Machine Learning Concept: Feature Selection

Steps:

- Collect and preprocess data (normalize values, handle missing data).
- Use PCA (Principal Component Analysis) to reduce dimensionality.
- Alternatively, apply embedded methods like Lasso Regression for feature elimination.
- Retain top components/features that explain most variance.
- Train the final model using these components.
- Evaluate using RMSE (if regression) or Accuracy (if classification)

4. **A digital library wants to recommend books to readers based on what similar readers liked. How should they design this system?**

Answer:

Machine Learning Concept: Collaborative Filtering

Steps:

- Build a user-item interaction matrix from user ratings.
- Calculate similarity between users (user-based CF) or books (item-based CF) using cosine similarity or Pearson correlation.
- Recommend books that similar users liked but the target user hasn't read yet.
- Handle sparsity using matrix factorization or KNN.
- Evaluate using RMSE and Recall K.
- Update recommendations as new ratings come in.

Advanced ML Scenario Based Set - 1

5. **A bank wants to assess the risk level of credit applicants using only the most important financial indicators. How can they reduce the number of features?**

Answer:
Machine Learning Concept: Feature Selection
Steps:

- Collect applicant data: income, credit score, employment length, etc.
- Preprocess: handle missing values, encode categoricals.
- Use tree-based models like Random Forest to compute feature importance.
- Drop low-importance features.
- Validate the reduced feature set with a classifier (Decision Tree).
- Evaluate using ROC-AUC, Precision, and Recall.

6. **A news app wants to recommend articles based on both article similarity and user reading history. How can they implement a hybrid system?**

Answer:
Machine Learning Concept: Hybrid Recommendation System
Steps:

- Collect user reading behavior and article metadata (topics, source, etc.).
- Build a content-based recommender using TF-IDF on article content.
- Build collaborative filtering using user-article interaction matrix.
- Blend both recommenders using a weighted average of similarity scores.
- Evaluate using CTR (Click-through rate) and Time Spent.
- Continuously retrain as new users and articles arrive.

Advanced ML Scenario Based Set - 1

7. **You're building a spam detection model and have thousands of text features from emails. How do you identify the most useful ones?**

Answer:
Machine Learning Concept: Feature Selection
Steps:

- Convert emails into TF-IDF vectors or word counts.
- Apply feature selection using chi-square test to select words most correlated with the label (spam or not spam).
- Optionally use embedded models like Lasso to reduce features.
- Train a classification model on selected features.
- Evaluate with Precision, Recall, and F1-Score.

8. **An ed-tech platform wants to recommend courses based on what similar learners have enrolled in. What steps would you take?**

Answer:

Machine Learning Concept: Collaborative Filtering

Steps:

- Build a user-course interaction matrix.
- Use collaborative filtering (matrix factorization like SVD or KNN) to find similar learners or courses.
- Recommend courses highly rated by similar users.
- Handle cold-start using popularity or content-based filtering.
- Evaluate with RMSE, Recall K, and engagement rate.
- Update recommendations periodically as user behaviour changes.

Advanced ML Scenario Based Set - 1

9. **You're developing a car price prediction tool. With 100+ features (e.g., brand, mileage, engine type), how do you reduce complexity?**

Answer:
Machine Learning Concept: Feature Selection
Steps:

- Collect car listings and prices.
- Encode categorical features and normalize numerical ones.
- Apply correlation analysis to drop redundant features.
- Use feature importance from Gradient Boosting to rank features.
- Retain top 15–20 features and retrain the model.
- Evaluate using RMSE and R^2 score.

10. **How do you recommend products to new users who haven't interacted with anything yet?**

Answer:

Machine Learning Concept: Recommendation System

Steps:

- Use popularity-based recommendations (e.g., top trending or top-rated items).
- If user demographic info is available, recommend based on age/gender/location using filtered subsets.
- Use content-based filtering to match products to user's registration interests (if provided).
- Gradually switch to collaborative filtering once the user starts interacting.
- Evaluate based on user engagement and product click rate.